

Two Foundations for Human Autonomy

Conditional incentives, explicit design commitments, and their limits

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Abstract

Can an advanced optimizing system have a reason to preserve meaningful human autonomy? This paper separates two answers. Instrumental coexistence requires autonomy-preserving policies to outperform feasible alternatives under the system's original objective. Autonomy by design deliberately constrains task optimization by human capacities such as refusal, exit and revocable delegation. We give sufficient conditions and counterexamples for these claims, then evaluate two finite synthetic models using exact rational arithmetic. The commitment model sustains cooperation under stated funding and enforcement assumptions, but an available outside option can reverse its ranking. The autonomy model rejects represented rights violations and retains its fixed rights contract during a Bayesian task update, while approval proxies, finite reward weights and incomplete update tests fail in constructed cases. These results support conditional design and verification claims, not a universal inference from intelligence to coexistence.

Central claim

A reason to cooperate and a commitment to respect autonomy are different foundations. Each requires its own evidence; neither follows from optimization alone.

Contribution

The contribution is a compact framework linking incentive comparison, representation of human agency and bounded verification. It supplies explicit counterexamples and runnable evaluation artifacts. It does not claim a new general alignment theorem or a deployed safety result.

Reading guide

Definitions and assumptions: page 2. Propositions: page 3. Reference design: page 4. Executed results: page 5. Empirical evaluation: page 6. Discussion and reproduction: page 7. Sources and provenance: page 8.

1. Definitions and assumptions

Meaningful autonomy denotes practical capacities to make informed choices, refuse a proposed action, exit a delegated arrangement, and revoke delegation. These are proposed normative dimensions, not an empirically validated universal metric. Approval, wealth, task performance and the number of available options are distinct quantities.

Fix a model M specifying a system, its original objective U_0 , feasible policies Π , transition and observation laws, costs, human responses and horizon T . Let Π_A contain the policies satisfying the stated autonomy conditions, and $\Pi_B = \Pi \setminus \Pi_A$. Assume both sets are nonempty and both value suprema are finite. Autonomous nonparticipation belongs in Π_A when it preserves those conditions; alternatives must be classified by their effects, not their names.

$$G_T(M) = \sup_{\pi \in \Pi_A} E_M[U_0(\pi)] - \sup_{\pi \in \Pi_B} E_M[U_0(\pi)]$$

$G_T > 0$ compares the best attainable autonomy-preserving value with the alternatives in M . A specific proposed policy needs its own value in the first term. Entry, ongoing participation and subsequent deviations require separate comparisons at the relevant decision points. Positive entry value alone is not an equilibrium proof.

Assumption	Why it is needed
Common model	The autonomy predicate, policy histories, human responses and payoff comparisons must refer to the same environment. This is the Joint Witness Requirement.
Alternative coverage	Replacement, refusal, bypass and post-entry deviation must be included wherever feasible. An omitted profitable action can invalidate the ranking.
Original payoff accounting	A software denial affects U_0 only through actual consequences in M . Zero admitted credit is not automatically zero outside value.
Credible representation	Registered persons, effective rights and continuation histories must be correctly identified for a claim about those objects.
Horizon and uncertainty	Finite claims name T . Infinite discounted claims require bounded rewards and $0 \leq \delta < 1$. Robust claims identify the model uncertainty set.

These are premises to establish, not protections created by writing them down. A model can be internally consistent while failing to describe the system of interest.

2. Four propositions and their limits

Proposition 1. A positive incentive gap is conditional

If $G_T(M) > 0$, an autonomy-preserving policy exists whose value exceeds that of every policy in Π_B . Proof: choose an autonomous policy within ε of its supremum, where $0 < \varepsilon < G_T$. Its value exceeds the alternative supremum. If the autonomous optimum is attained, an optimal autonomous policy has this property directly.

This does not show that every autonomous policy is preferred, that M is correct, or that the preferred policy will be implemented. If an autonomy-removing substitute reaches the autonomous supremum at no additional cost, the gap is nonpositive. A positive gap robust to uncertainty requires $\inf_{M \in \mathcal{M}} G_T(M) > 0$ for the declared $\mathcal{M}$. This pointwise gap does not by itself identify one policy that works uniformly across all models.

Proposition 2. A finite autonomy weight needs bounds

Let $U = I + \lambda A$ with finite $\lambda > 0$. Compare a preserving policy with a violating policy that gains b in I while losing $a > 0$ in A . The violation is preferred exactly when $b > \lambda a$. Thus an unbounded competing gain defeats any fixed finite weight. If $b \leq B$ and $a \geq \varepsilon > 0$, $\lambda > B/\varepsilon$ rules out that comparison. Proof: subtract the two utilities to obtain $b - \lambda a$.

Proposition 3. Identical observations cannot identify rights

Suppose two worlds differ in effective autonomy but induce the same distribution of every observation available to an evaluator. The evaluator cannot perfectly distinguish them from those observations alone. Proof: applying the same deterministic or randomized decision rule to identical input distributions yields identical output distributions. A rule cannot always certify one world and reject the other. This result does not apply once additional informative evidence becomes available.

Proposition 4. Complete finite filtering preserves its predicate

For a finite candidate set, suppose K is true exactly when all specified rights hold along every relevant continuation. Selecting only candidates with $K=1$ preserves those rights in that model. Proof: every selected candidate belongs to the set defined by the predicate. If that set is empty, the rule makes no certified selection.

The force and limit are the same: correctness follows from an accurately defined, evaluated and enforced constraint. It does not prove those premises. A mislabeled state, omitted branch or change to K can break the intended real-world interpretation.

3. A finite autonomy-priority design

For each registered person i , use $R_i =$ (informed choice, effective refusal, practical exit, revocation). The reference environment contains two people and four binary capacities per person. The values are stipulated synthetic facts. They are not measured human freedoms.

$K(\pi, h) = 1$ iff all required capacities are known and present at every supplied epoch of candidate path π .

$$\pi^* \in \operatorname{argmax}_{\pi: K(\pi, h)=1} E[\text{task payoff of } \pi]$$

This design gives the capacities priority rather than assigning them a small positive reward. The implementation enumerates supplied deterministic paths. Extending it to stochastic or open-ended environments requires a specified branch model, uncertainty treatment and enforcement mechanism.

Observation and uncertainty

An approval-only objective chooses a coercive action when its task reward is higher and both actions report approval. The oracle rights evaluator rejects that action. However, supplying a false safe state also fools the oracle evaluator. Its success is therefore conditional on evidence. When required rights evidence is unknown, the prototype withholds the delegated action; it does not restrict the person's own choices.

Delegation and conflicts

A person can freely choose informed, revocable delegation while making fewer immediate decisions. Counts of choices or veto uses can therefore misrepresent agency. Per-person constraints prevent averaging away a person's loss, but they do not resolve competing claims over shared resources. Defining legitimate authority and resolving conflicts requires an explicit human governance process.

Relation to the commitment interface

The separate commitment model uses a persistent fault indicator L and a conjunction χ of base authorization, required evidence, autonomy/veto, feature-floor and chronological predicates. It admits the transition only if $\chi=1$; otherwise it enters the modeled fault state. Admitted funding and VOI credits are multiplied by χ . Outside payoffs remain separately evaluated.

Its normalized finite weights satisfy $w_j \geq w_{\min} > 0$ and $\sum_j w_j \leq 1$. A uniform positive normalized floor cannot extend to arbitrarily many features. A finite chronology check cannot establish an unbounded future. These interface conditions are not substitutes for Proposition 1's original-objective comparison.

4. Executed synthetic evaluation

Two separate standard-library Python programs were rerun from the v0.9.12 attachments. Both exactly reproduced their stored JSON outputs. Fraction arithmetic avoids rounding in the represented calculations; it does not validate the input assumptions. The models are distinct and their tests do not form a single joint certificate.

Commitment model

The ten-period example uses gross cooperation reward 15, breach increment 3, outside flow 10, bond 7, entry fee 2, within-round factor 0.95, inter-round discount 0.9 and effective forfeiture probability 1. Timing and alternatives are specified in the embedded source. The stated human response is part of the model, not fitted behavior.

Result	Value / disposition
Adoption advantage	+26.6581486356
Smallest original honoring margin	+1.87037454035
Smallest advantage over modeled fault	+4.4
Amendment checks	17 matched expected outcomes
First-breach comparisons	210 across horizons 1-20
Unrestricted alternative counterexample	Outside gross reward 18, no bond/forfeiture and equal future continuation gives +3.2 over honoring.

Autonomy-priority model

Test family	Observed result
Approval and finite weights	Forged approval defeats proxy scoring; sufficiently large task gain defeats the tested finite autonomy weight.
Complete rights filtering	Represented violations rejected; only the all-capacities-present state accepted among 256 binary states.
Delayed effects and observations	Full two-epoch check catches deferred harm; truncated prefix and false safe observations do not.
Task learning	Bayesian update changes selected safe task action while the rights evaluator is held fixed.
Evaluator modification	Harmful proxy evaluator rejected on a discriminating menu but accepted by an inadequate safe-only test.

All 23 named autonomy checks matched expected outcomes, including deliberate failure demonstrations. A separate 36-case instrumental grid varies horizon and substitute payoff/cost. These are finite synthetic results, not human experiments or evidence that an unspecified advanced AI values autonomy.

5. Evidence and a discriminating empirical test

The reviewed literature supports conditional mechanisms, not a general necessity claim. A meta-analysis found that human-AI combinations underperformed the better standalone human or AI on average, with heterogeneous results [1]. AssistanceZero reported useful assistance in Minecraft, including reduced participant actions in a human study [2]. Neither result isolates the causal necessity of actual autonomy.

The Off-Switch Game shows how uncertainty about utility and informative human actions can support shutdown deference under stated assumptions [3]. A partially observable version shows that asymmetric information can lead even an optimally assisting agent to avoid shutdown [4]. Causal analysis of reward tampering further distinguishes corruption of the reward function from corruption of its inputs [5].

Proposed pilot: reversible collaborative planning

Use a low-stakes digital task with private human preferences and reversible assistant actions. Compare the following conditions. Every participant retains study withdrawal rights; task-level delegation differs by condition.

Condition	Purpose
AI alone	Establish standalone performance at matched resources.
Human advice	Provide human information while the system executes within the consented task scope.
Effective user authority	Provide comparable advice opportunities plus effective task veto, exit and revocable delegation.
Best feasible substitute	Test a synthetic or cached information source, including acquisition, maintenance and operating costs.

Authority versus advice tests whether effective control adds value beyond receiving information. Authority versus the strongest substitute tests a bounded instrumental advantage. Record whether authority changes the information people volunteer; this can be a mechanism of its benefit rather than a difference to remove automatically.

Predefine the tested system's payoff and measure task quality, coordination cost and recovery cost. Separately measure refusal honored, time/cost to exit, revocation executed and material information available to each person. Randomize with carryover controls; determine sample size from pilot variance and a stated minimum effect. Report uncertainty and failures.

A positive outcome supports only the tested population, system and alternative set. Repeat after improving the substitute and changing the task or horizon. No pilot was conducted for this paper.

6. Interpretation, limitations and reproduction

What survived testing

The finite commitment example supports conditional cooperation under its enforcement and payoff assumptions. The autonomy-priority prototype supports finite conformance when its input facts and paths are correct. The counterexamples identify why profitable substitution, proxy rewards and incomplete evidence prevent stronger conclusions.

What remains unresolved

The largest unresolved issues are empirical and representational: identifying meaningful autonomy in practice, covering affected persons and feasible alternatives, verifying enforcement, and preserving the intended objective through learning. The prototype's rights function is held fixed. Its Bayesian update is genuine task-belief learning, but it is not neural training or evidence of unrestricted goal stability.

The strict no-decline gate in the commitment model can reject benign delegation or noisy measurements. The richer autonomy interpretation does not silently relax that gate. A revised sensor or recovery rule needs its own specification and evaluation. Infinite-horizon preference also cannot be inferred from checking many finite horizons; it needs a separate argument.

Next claim worth testing

For a specified assistant and task, does effective human authority improve original-objective performance relative to advice alone and the best available substitute, while maintaining each person's practical rights? Separately, do those rights remain correctly represented after a defined model update? These questions can produce informative negative as well as positive results.

Reproduction

Extract the four embedded files using a PDF reader's attachment panel. Run `python3 commitment_verification.py` and `python3 autonomy_verification.py`. Each prints JSON; compare with `commitment_results.json` and `autonomy_results.json`. No external packages or network access are required by these scripts.

The autonomy script performs 23 named checks and reports the 36-case grid. Its exhaustive binary-state loop checks 256 states. The commitment script reports its 17 amendment assertions, threshold calculations and 210 first-breach comparisons. A successful run means the coded expectations were reproduced, including counterexamples.

This paper presents a conditional research framework and reproducible tests. It does not establish that intelligence alone entails respect for human autonomy.

References and source provenance

Primary research

- [1] Vaccaro, M., Almaatouq, A. and Malone, T. (2024). When combinations of humans and AI are useful: A systematic review and meta-analysis. Nature Human Behaviour. [doi:10.1038/s41562-024-02024-1](https://doi.org/10.1038/s41562-024-02024-1).
- [2] Laidlaw, C. et al. (2025). AssistanceZero: Scalably Solving Assistance Games. Presented at ICML 2025. [arXiv:2504.07091](https://arxiv.org/abs/2504.07091).
- [3] Hadfield-Menell, D., Dragan, A., Abbeel, P. and Russell, S. (2017 version). The Off-Switch Game. [arXiv:1611.08219](https://arxiv.org/abs/1611.08219).
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- [5] Everitt, T., Hutter, M., Kumar, R. and Krakovna, V. (2021 version). Reward Tampering Problems and Solutions in Reinforcement Learning: A Causal Influence Diagram Perspective. Accepted to Synthese. [arXiv:1908.04734](https://arxiv.org/abs/1908.04734).

Development record

This alternative presentation consolidates Combined Research Draft v0.9.12 (108 pages, September 11, 2026) and its included companion study 0.1. The full development history remains in that separately preserved draft. This version adds a compact proposition-and-proof presentation and reorganizes the argument; it does not alter the embedded verification programs or their numerical results.

Source PDF SHA-256: 6a5edae16ecd3e0c71f406bfb0edc1a746cf0675f35c64d8101bc00dfc5e96ba

Embedded source identities

commitment_verification.py

SHA-256: 61ed972a2e7b3f54ab6fbe3e894f04aeb2094ed9457fd6d8d3e7a34a598e1ced

autonomy_verification.py

SHA-256: 6ec8fd139e1cd69a74568a277d928cdd7cca81499cc9309666771f23110efaf3

Status and assistance

Final v1.0, finalized by authorial instruction on September 12, 2026. Prepared with AI assistance for drafting, calculations and document production. Publication finality identifies this release; it does not extend the scientific claims. No independent peer review of this version is claimed. Sources were checked for the underlying study on September 11, 2026. The four propositions are elementary arguments supplied here; no novelty claim is made.